



Explainable Artificial Intelligence

(XAI)



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ΟΙΚΟΔΟΜΗ ΣΗΜΕΙΩΣΗ ΜΗΜΑΡΟ ΟΜΟΘΕΤΟΝΤΕΣ ΠΟΛΥΕΠΙΣΤΗΜΟΝΕΣ - ΘΕΟΡΟΙ

Chapitre 6

Méthodes Locales Post-Hoc Counterfactual explanations



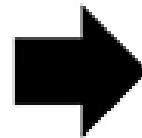


Explication contrefactuelle

Comprendre et implémenter les contrefactuels

Données originales

Glucose	BMI	Exercise (mins)	Diabetic
175	22	19	False



Counterfactual 1

Glucose	BMI	Exercise (mins)	Diabetic
195	22	19	True

Counterfactual 2

Glucose	BMI	Exercise (mins)	Diabetic
95	22	5	True

Counterfactual 3

Glucose	BMI	Exercise (mins)	Diabetic
95	29	19	True

Contrefactuel: C'est la plus petite variation d'une caractéristique qui modifie la prédiction vers une autre sortie.



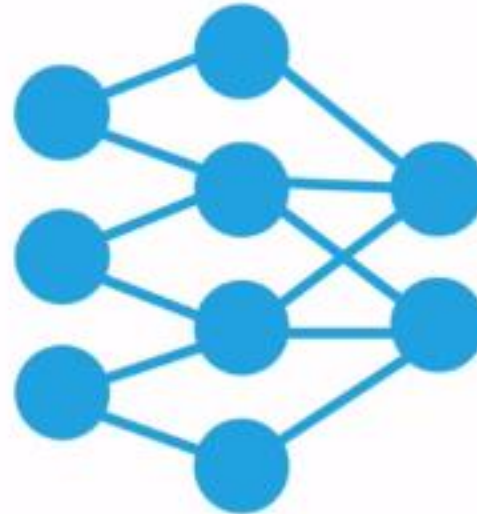
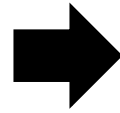


Explication contrefactuelle

Signification des contrefactuels

SARA

Glucose	BMI	Exercise (mins)	Calories
195	22	19	2000



Complexe ML model



Diabetes probability





Explication contrefactuelle

Comment je peux réduire ma probabilité

SARA

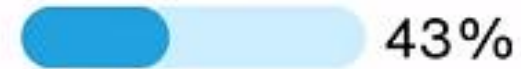
Glucose	BMI	Exercise (mins)	Calories
195	22	19	2000



Glucose	BMI	Exercise (mins)	Calories
195	22	19	1500



Probabilité de diabète



Probabilité de diabète





Explication contrefactuelle

Comment je peux réduire ma probabilité

SARA

Glucose	BMI	Exercise (mins)	Calories
195	22	19	2000



Glucose	BMI	Exercise (mins)	Calories
195	22	19	1500

Diabétique

Non diabétique

Quelle est la plus petite modification que Sara peut apporter à ses caractéristiques (par exemple son taux de glucose, son IBM, etc.) pour que la prédiction du modèle passe de “diabétique” à “non diabétique” ?





Explication contrefactuelle

Comprendre et implémenter les contrefactuels

Jeu de données de prédiction des accidents vasculaires cérébraux (AVC)

healthcare-dataset-stroke-data.csv (316.97 kB)  

Detail Compact Column 10 of 12 columns 

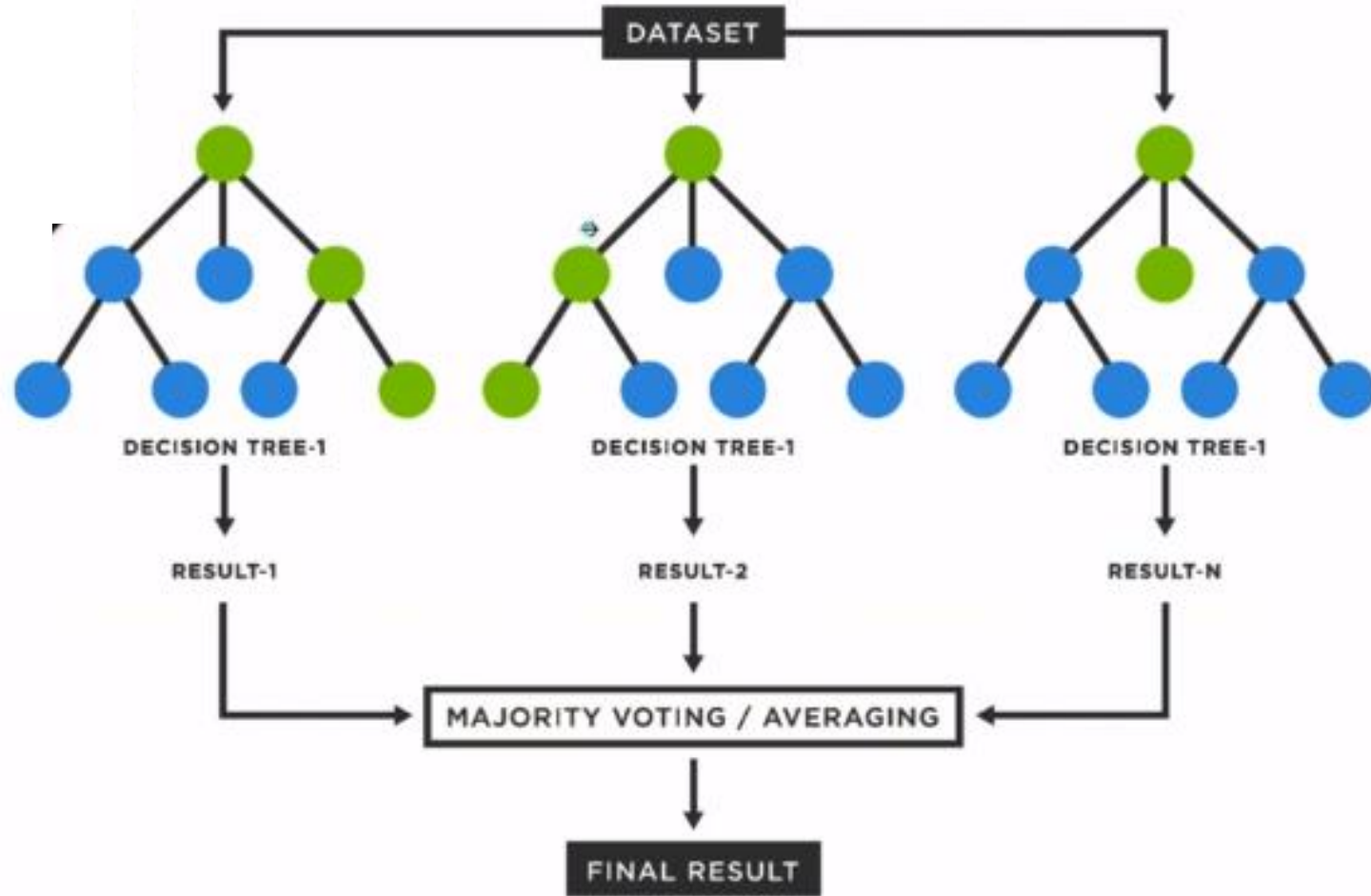
 id 	 gender 	 age 	 hypertension 	 heart_disease 	 ever_married 
Unique id	Gender	Age	Hypertension binary feature	Heart disease binary feature	Has the person been married
 67 72.9k	Female 58% Male 41% Other (1) 0%	 0.08 82	 0 1	 0 1	
53882	Male	74	1	1	Yes
10434	Female	69	0	0	No
27419	Female	59	0	0	Yes
60491	Female	78	0	0	Yes
12189	Female	81	1	0	Yes
12095	Female	61	0	1	Yes
12175	Female	54	0	0	Yes





Explication contrefactuelle

Model : RANDOM FOREST CLASSIFIER





Explication contrefactuelle

Model : RANDOM FOREST CLASSIFIER

Charger le jeu de
données sur les
AVC

Entraîner un
Random Forest

Charger DiCE
(Explications
contrefactuelles
diversifiées)

Analyser les
caractéristiques qui
peuvent être modifiées
pour inverser la
prédiction du modèle





Explication contrefactuelle

Exemple

Install libraries

```
# pip install torch
```

```
# pip install dice-ml
```

Import libraries

```
# %% Imports
from torch.utils.data import DataLoader
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import f1_score, accuracy_score

import pandas as pd
from sklearn.model_selection import train_test_split
from imblearn.over_sampling import SMOTE
```

```
# path = '/content/data/healthcare-dataset-stroke-data.csv'
```





Explication contrefactuelle

Exemple

Load the data

```
# %% Custom DataLoader
class CustomDataLoader:
    def __init__(self, filepath):
        self.filepath = filepath
        self.data = None

    def load_dataset(self):
        self.data = pd.read_csv(self.filepath)

    def preprocess_data(self):
        # Implement your preprocessing here
        self.data.dropna(inplace=True)
        self.data = pd.get_dummies(self.data)

    def get_data_split(self, test_size=0.2, random_state=42):
        X = self.data.drop('stroke', axis=1)
        y = self.data['stroke']
        return train_test_split(X, y, test_size=test_size, random_state=random_state)

    def oversample(self, X_train, y_train):
        smote = SMOTE(random_state=42)
        X_res, y_res = smote.fit_resample(X_train, y_train)
```





Explication contrefactuelle

Exemple

Train-Test data split

```
# Split the data for evaluation
X_train, X_test, y_train, y_test = data_loader.get_data_split()

# Oversample the train data
X_train, y_train = data_loader.oversample(X_train, y_train)
```

```
y_test = y_test.reset_index(drop=True)
X_test = X_test.reset_index(drop=True)
y_train = y_train.reset_index(drop=True)
X_train = X_train.reset_index(drop=True)
```

```
(3927, 22)
(982, 22)
(7542, 22)
(982, 22)
```





Explication contrefactuelle

Exemple

Random Forest Classifier

```
# %% Fit blackbox model
rf = RandomForestClassifier()
rf.fit(X_train, y_train)
y_pred = rf.predict(X_test)
print(f"F1 Score {f1_score(y_test, y_pred, average='macro')}")
print(f"Accuracy {accuracy_score(y_test, y_pred)}")
```



```
# Convert y_test and y_pred to pandas Series
```

```
y_test_series = pd.Series(y_test)
```

```
y_pred_series = pd.Series(y_pred)
```

```
# Get indices where y_test and y_pred are 1
```

```
test_indices = y_test_series[y_test_series == 1].index.tolist()
```

```
pred_indices = y_pred_series[y_pred_series == 1].index.tolist()
```

```
print("Test indices:", test_indices)
```

```
print("Prediction indices:", pred_indices)
```

```
... Test indices: [11, 30, 35, 62, 73, 110, 113, 122, 143, 166, 167, 198, 229, 238, 274, 277, 299, 312, 327, 336, 362, 388, 426, 434, 475, 488, 499, 538, 57]
Prediction indices: [131, 254, 288, 379, 613, 679, 704, 971]
```





Explication contrefactuelle

Exemple

Create Counterfactual Explanations

```
# %% Create diverse counterfactual explanations
import dice_ml

# Dataset
data_dice = dice_ml.Data(dataframe=data_loader.data,
                          # For perturbation strategy
                          continuous_features=['age',
                                              'avg_glucose_level',
                                              'bmi'],
                          outcome_name='stroke')
```





Explication contrefactuelle

Exemple

Creating the Data and Model Objects for DiCE: (Diverse Counterfactual Explanations)

```
# Model
rf_dice = dice_ml.Model(model=rf,
                        # There exist backends for tf, torch, ...
                        backend="sklearn")
explainer = dice_ml.Dice(data_dice,
                        rf_dice,
                        # Random sampling, genetic algorithm, kd-tree,...
                        method="random")
```





Explication contrefactuelle

Exemple

Generating and Visualizing Counterfactual Explanations:

Double-click (or enter) to edit

```
# %% Create explanation
# Generate CF based on the blackbox model
input_datapoint = X_test[10:11]

cf = explainer.generate_counterfactuals(input_datapoint,
                                       total_CFs=3,
                                       desired_class="opposite")

print(X_test[0:1])

# Visualize it
# cf.visualize_as_dataframe(show_only_changes=False)

cf.visualize_as_dataframe(show_only_changes=True)
```





Explication contrefactuelle

Exemple

Creating Feasible (Conditional) Counterfactuals

```
# Get indices where age is above 70
indices_above_70 = X_test[X_test['age'] > 70].index.tolist()

print("Indices of people whose age is above 70:", indices_above_70)
```

```
Indices of people whose age is above 70: [0, 10, 22, 39, 56, 62, 63, 76, 79, 88, 106, 110, 113, 131, 139, 142, 143, 147, 183, 195, 202, 205, 206, 208, 2
```





Explication contrefactuelle

Exemple

```
# %% Create feasible (conditional) Counterfactuals
features_to_vary=['avg_glucose_level',
                  'bmi',
                  'smoking_status_smokes']
permitted_range={'avg_glucose_level':[40,300],
                  'bmi':[15, 45]}

i = 139

input_datapoint2 = X_test[i:i+1]

print("Label of test data: ", y_test[i])
print(input_datapoint2.to_string(index=False))

# Now generating explanations using the new feature weights

cf = explainer.generate_counterfactuals(input_datapoint2,
                                       total_CFs=10,
                                       desired_class="opposite",
                                       permitted_range=permitted_range,
                                       features_to_vary=features_to_vary)
```





Explication contrefactuelle

Exemple

```
# Visualize it
cf.visualize_as_dataframe(show_only_changes=True)
```

Label of test data: 0

	id	age	hypertension	heart_disease	avg_glucose_level	bmi	gender_Female	gender_Male	gender_Other	ever_married_No	ever_married_Yes	work_type_
44873	81.0	0	0	125.2	40.0	1	0	0	0	0	1	

100%|██████████| 1/1 [00:00<00:00, 2.82it/s]Query instance (original outcome : 0)

	id	age	hypertension	heart_disease	avg_glucose_level	bmi	gender_Female	gender_Male	gender_Other	ever_married_No	...	work_type_Private	work_type_Public
0	44873	81.0	0	0	125.199997	40.0	1	0	0	0	...	0	0

1 rows x 23 columns





Explication contrefactuelle

Exemple

Diverse Counterfactual set (new outcome: 1)

	id	age	hypertension	heart_disease	avg_glucose_level	bmi	gender_Female	gender_Male	gender_Other	ever_married_No	...	work_type_Private	work_o
0	-	-	-	-	-	-	0.0	-	-	-	...	-	
1	-	-	-	-	-	-	-	-	-	-	...	-	
2	-	-	-	-	-	-	-	-	-	-	...	-	
3	-	-	-	-	-	-	0.0	-	-	-	...	-	
4	-	-	-	-	-	-	-	-	-	-	...	-	
5	-	-	-	-	-	18.9	-	-	-	-	...	-	
6	-	-	-	-	-	-	0.0	-	-	-	...	-	
7	-	-	-	-	-	-	0.0	-	-	-	...	-	
8	-	-	-	-	257.4	-	0.0	-	-	-	...	-	
9	-	-	-	-	-	-	-	-	-	-	...	-	

10 rows x 23 columns





Explication contrefactuelle

Exemple

```
print(y_test[i])
```

```
0
```

Start coding or [generate](#) with AI.

100%|██████████| 1/1 [00:00<00:00, 1.85it/s]No Counterfactuals found for the given configuration, perhaps try with different parameters... ; total time taken: 00 min 00 se

UserConfigValidationException Traceback (most recent call last)

<ipython-input-158-21742cea308a> in <cell line: 3>()

```
1 # Now generating explanations using the new feature weights
```

```
2
```

```
----> 3 cf = explainer.generate_counterfactuals(input_datapoint2,
```

```
4                                     total_CFs=2,
```

```
5                                     desired_class="opposite",
```

↕ 1 frames

/usr/local/lib/python3.10/dist-packages/dice_ml/explainer_interfaces/explainer_base.py in _check_any_counterfactuals_computed(self, cf_examples_arr)

```
880         break
```

```
881     if no_cf_generated:
```

```
--> 882         raise UserConfigValidationException(
```

```
883             "No counterfactuals found for any of the query points! Kindly check your configuration.")
```

```
884
```

UserConfigValidationException: No counterfactuals found for any of the query points! Kindly check your configuration.





Explication contrefactuelle

Comprendre et implémenter les contrefactuels

Exemple Code:

<https://colab.research.google.com/drive/1zttcqlvmCqjPy33UG4ixMAMZ6P3vubJG?usp=sharing>

